Physics - Graduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Light of wavelength 400 nm is incident upon lithium (phi = 2.93 eV). Calculate the stopping potential in V.
   - (a) 0.87
   - (b) -0.23
   - (c) 0.07
   - (d) 0.57
   - (e) 0.97

2. A voltage will be induced in a wire loop when the magnetic field within that loop
   - (a) is amplified
   - (b) converts to magnetic energy
   - (c) is parallel to the electric field
   - (d) changes
   - (e) is at right angles to the electric field

3. Calculate the binding energy of the hydrogen atom in its ground state.
   - (a) 14.1 eV
   - (b) 10.6 eV
   - (c) 11.2 eV
   - (d) 17.0 eV
   - (e) 15.3 eV

4. $B^{1}0$ is bombarded with neutrons, and α particles are observed to be emitted. What is the residual nucleus?
   - (a) $^{8}\_4Be$
   - (a) $^{1}3\_6C$
   - (a) $^{7}\_4Be$
   - (a) $^{6}\_3Li$
   - (a) $^{5}\_2He$

5. The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons

(a) an average of 10 times, with an rms deviation of about 5

(b) an average of 10 times, with an rms deviation of about 3

(c) an average of 10 times, with an rms deviation of about 0.1

(d) an average of 10 times, with an rms deviation of about 1

(e) an average of 10 times, with an rms deviation of about 6

True / False (10 marks)

Answer True or False

6. True or False: The resistance of an electric toaster operating at 5 amperes on a 120-volt circuit is 28 ohms, calculated using Ohm's law.

7. True or False: In a spring-block oscillator, the maximum speed of the block is inversely proportional to the amplitude of oscillation.

8. True or False: The refracting power of a combination of a concave lens with focal length 12 cm and a convex lens with focal length 7.5 cm is -5 diopters.

9. True or False: A step-up transformer can increase both the current and the power delivered to the load without any losses.

10. True or False: An object with a mass of 1 gram experiences an acceleration of 500 cm/s^2 when subjected to a force of 2000 dynes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between capacitance, potential difference, and charge in capacitors. Given a capacitor with a capacitance of 2 mF charged to 5 V, calculate the charge on the positive plate and analyze the implications of your result.

12. Discuss the conditions under which Avogadro's number of translational states becomes available for O_2 in a confined volume of 1000 cm^3, and determine the corresponding temperature.

End of Paper